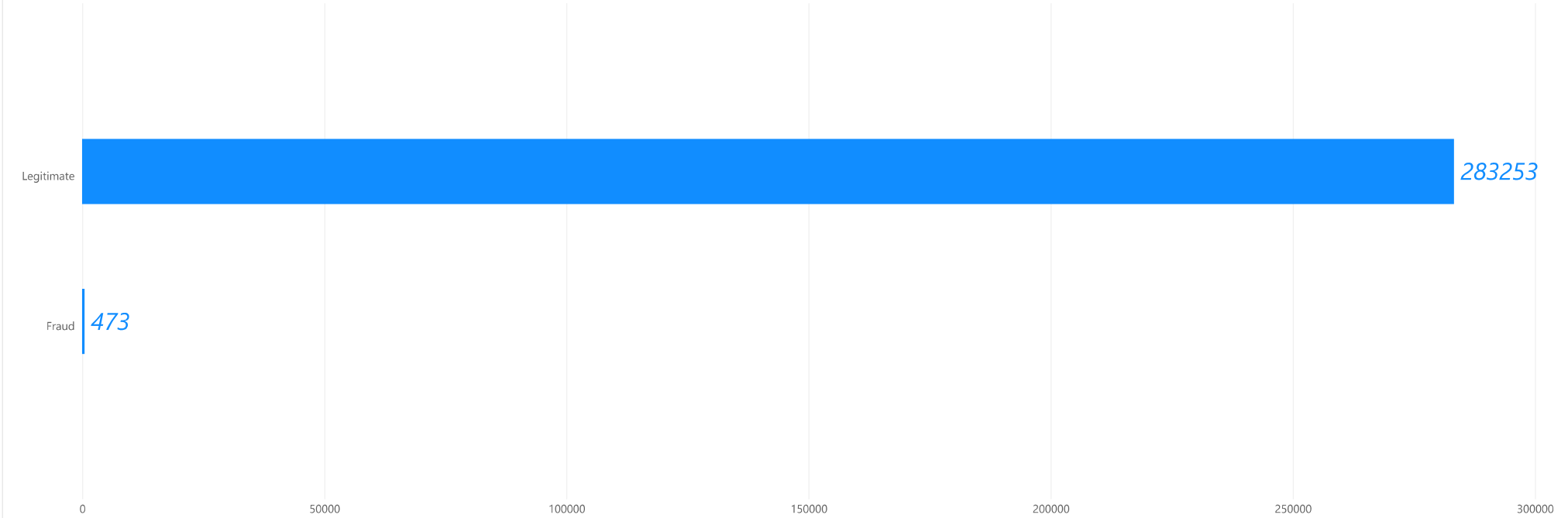
FRAUDSHIELD AI | Credit Card Fraud Intelligence

Total Transactions	Fraud Cases	Fraud Rate	Fraud Caught	Model Precision	Model Recall
284K	473	0.17%	71	92.21%	74.74%

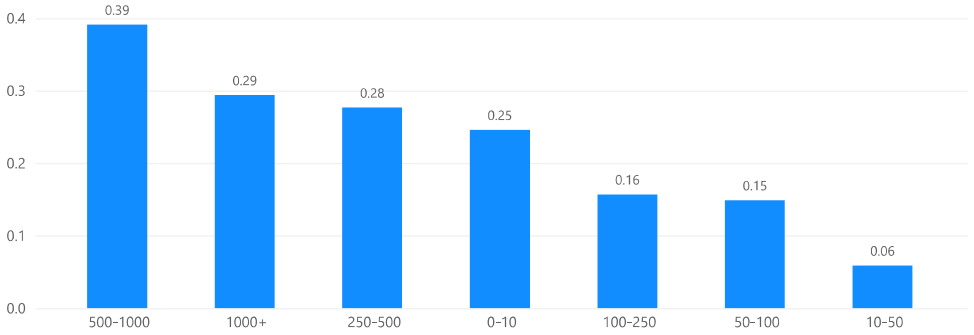
Class Imbalance: Legitimate vs Fraud Transactions



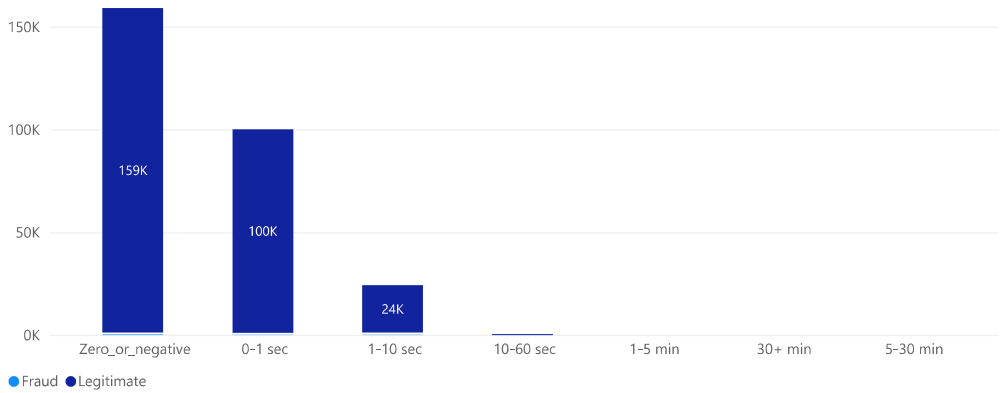
Source: Kaggle Credit Card Fraud Detection | Model metrics evaluated on held-out test split.

Quick measure

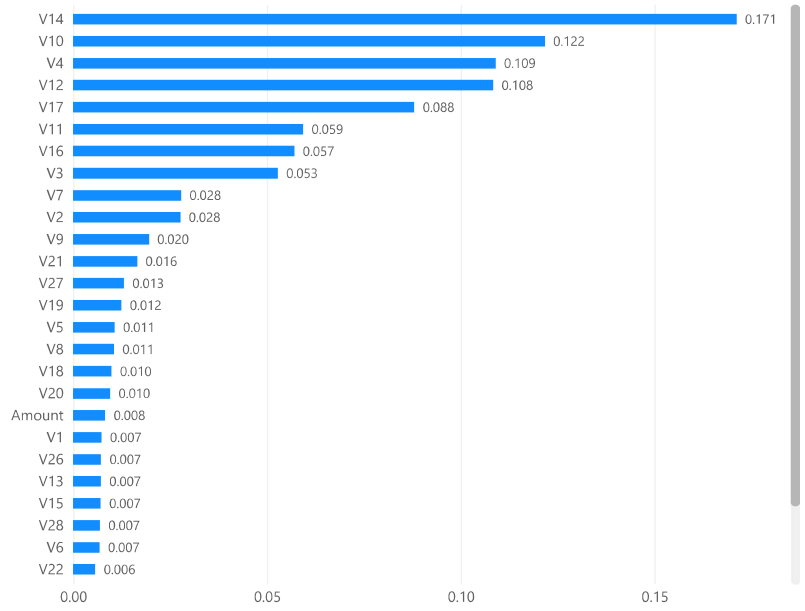
Fraud rate by transaction amount



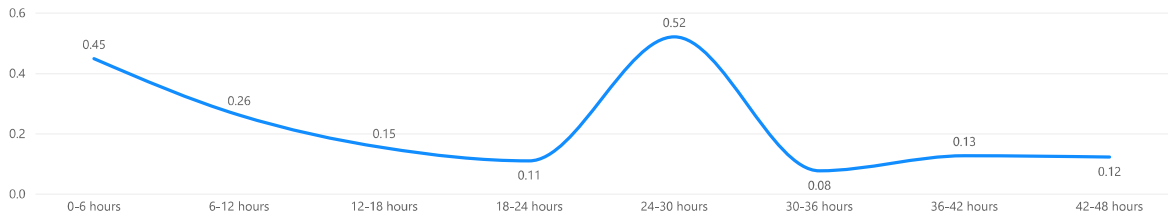
Time-gap distribution



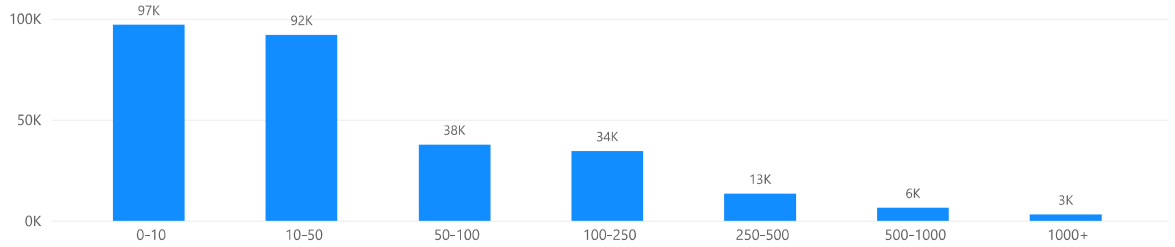
Feature importance



Fraud rate by elapsed-time interval



Transaction volume by amount



Quick measure

FRAUDSHIELD AI | Credit Card Fraud Intelligence

Context rules raise priority but cannot trigger alerts alone. Each has a fraud rate of roughly 0.5% or lower, so most of their alerts would be false. The model score is the primary trigger.

Fraud Caught	Fraud Missed	False Alerts	Total Alerts
71	24	6	77

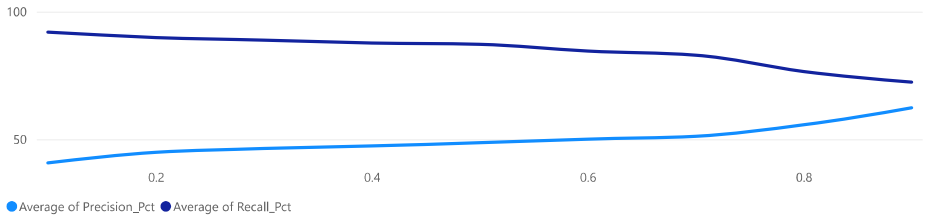
"Random Forest, threshold 0.3, held-out test set (56,746 transactions, 95 frauds)"

Model

Logistic Regression | Random Forest

Model	Sum of Threshold	Sum of TP	Sum of FN	Sum of FP	Sum of Alerts	Sum of Precision_Pct	Sum of Recall_Pct
Logistic Regression	4.50	772	74	23338	24110	71.91	821.28
Random Forest	4.50	661	185	69	730	822.77	703.19
Total	**9.00**	**1433**	**259**	**23407**	**24840**	**894.68**	**1,524.47**

Precision vs Recall by Alert Threshold (validation set)



Confusion Matrix (test set)

Actual_Class	Fraud	Legitimate	Total
Fraud	71	24	**95**
Legitimate	6	56645	**56651**
Total	**77**	**56669**	**56746**

- **Score ≥ 0.8:** block and call the customer. 60 frauds caught, only 2 false alerts, 96.8% precision.
- **Score 0.3–0.8:** step-up authentication or manual review. At 0.3: 81 frauds caught, 9 false alerts, 90% precision, 86% recall.
- **Score 0.1–0.3:** monitor silently. At 0.1: 83 caught, 20 false alerts.
- **Score < 0.1:** allow.

Risk signals that raise priority

- Early windows (0–6 h and 24–30 h after dataset start): about 0.48% fraud rate vs 0.17% overall.
- $500–1000 transactions: about 0.39% fraud rate, the highest amount band.
- Micro-transactions under $10: 238 of 473 frauds, so half of all fraud.
- Extreme values in V14, V10, V4, V12 and V17 push a score up.

Source: Kaggle Credit Card Fraud Detection | Model metrics evaluated on held-out test split.

Quick measure